National and international collaboration to facilitate human genomics data sharing in Australia:

The Human Genomes Platform Project

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What is the HGPP?

The Human Genomes Platforms Project (HGPP) is a nationally funded initiative that aims to leverage best practice technologies and global standards to accelerate FAIR human genomics data sharing in Australia. Involving Australia's largest human genomic sequence generators, along with national computing infrastructure partners, the HGPP will break down silos and facilitate the deployment of a services toolbox across Australian genomic research institutes.

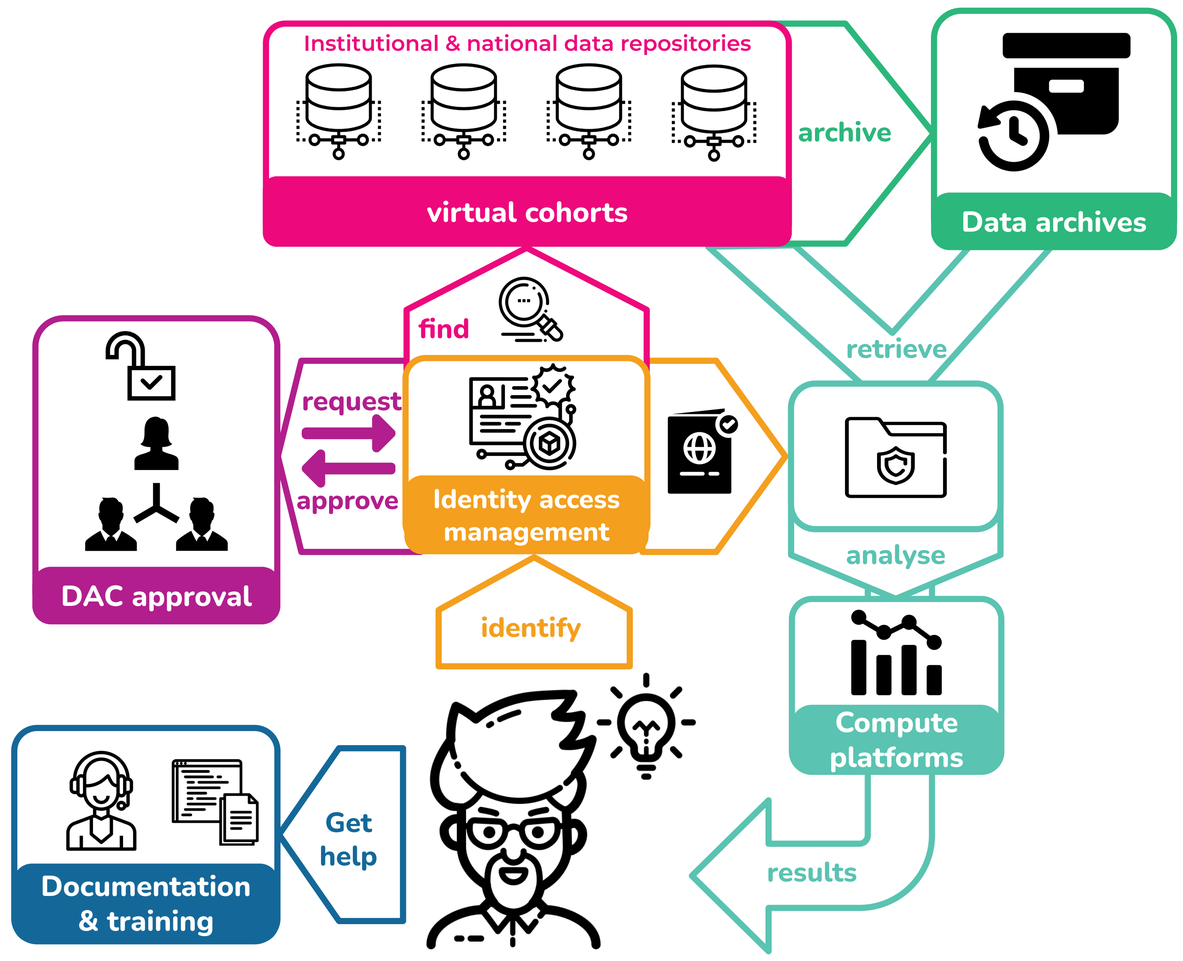
Virtual Cohorts

Currently, searching across national data repositories is challenging due to limited data sharing infrastructure support. The virtual cohorts sub-project is enabling cross-repository searching by facilitating implementation of data commons and lighting a series of Beacon v.2 instances, to allow cross-institution querying via a Beacon network.

DAC Automation

The current Data Access Committee (DAC) request and approval process is manual across partner institutes. As the number and size of datasets grow; tracking, auditing and managing this process will become increasingly challenging. The DAC Automation sub-project is investigating tools to facilitate the DAC approval and request process, including:

- Resource Entitlement Management System (**REMS**)
- **CILogon**
- Data Use Oversight System (**DUOS**)



[Meta]data Archiving

Australian researchers need to archive their human genomics research data in a controlled access repository but encounter challenges with submission to the Central EGA. The Data and Metadata Archiving sub-project is investigating ways of improving the submission process for Australian researchers and investigating the requirements for setting up a Local EGA node. We are piloting a test instance on National Compute Infrastructure and will produce a detailed feasibility report for this important national service.

Documentation, Communications & Training

The documentation, communications and training sub-project is ensuring all newly implemented technologies are adequately documented and providing support to deliver training in their uptake and adoption. Reporting outputs will be uploaded to Zenodo to enable easy access. Webinars and news articles detailing outputs and findings will be circulated to the broader community.

Federated Identity & Access Management

In a similar vein to the **ELIXIR** and EOSC-Life Life Science AAI systems, the Federated IAM sub-project is exploring and implementing systems that can be used across Australian institutes to confidently ascertain a user's identity and manage their rights to access national genomics resources and services. We are currently assessing the suitability of CILogon and the best way of establishing a National Researcher identity.

The HGPP received investment from the NCRIS-enabled ARDC infrastructure under investment identifier https://doi.org/10.47486/PL032 as well as being funded through BioPlatforms Australia. Contributions are also made from each partner organisation: QIMR Berghofer Medical Research Institute, The University of Melbourne Centre for Cancer Research, Garvan Institute for Medical Research, ZERO Childhood Cancer & the Children's Cancer Institute, Australian Genomics, Melbourne Genomics Health Alliance, National Computational Infrastructure and the Australian Access Federation.

Icons from the Noun Project: Computing, search, cloud by Flatart, database by Start Up Graphic Design, Analyse by Taylan Sentürk, computing by Phonlaphat Thongsriphong, identified by Tippawan Sookruay, group by Gregor Cresnar, group by Oksana Latysheva, Data File by Blangcon, Unlock by Arthur Shlain, archive by Adrien Coquet, support by Komkrit Noenpoempisut, documentation by lastspark, Scientist by Maxim Kulikov, Immigration Approval by Ary Prasetyo

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